Exoplanet Database Crossmatching applied to HPIC and LIFEcat for LIFE target selection

Joshua Dreier
jodreier@ethz.ch

September 14, 2026

Abstract

Target-selection catalogs for missions such as LIFE and NASA's Habitable Worlds Observatory benefit from knowing which already-known exoplanets orbit their candidate stars. We present an open-source, catalog-agnostic Python package that crossmatches a stellar catalog against a planetary catalog and computes missing, physically inferable parameters. Applied to HPICxLIFEcat, a stellar catalog extending HPIC with the non-binary M dwarfs of LIFEcat, the pipeline combines identifier-based matching, using alternate identifiers from SIMBAD and Exo-MerCat, with a coordinate-based matching fallback, against both the NASA Exoplanet Archive and Exo-MerCat. We calibrate the coordinate-matching thresholds, characterize the resulting matching yield, and use the enriched catalog to identify confirmed and candidate rocky, temperate planets around HPICxLIFEcat stars, providing a target list relevant to LIFE's characterization phase.

1 Introduction

1.1 Context

LIFE (*Large Interferometer for Exoplanets* [1]) is a mission concept consisting of four free-flying telescopes operating together as a nulling interferometer in space. It is proposed to directly image the thermal emission of low-mass temperate exoplanets in the mid-infrared. The mission is split into two phases: a search phase and a characterization phase. The mission notably shares the science goal of finding and characterizing habitable planets with NASA's proposed HWO (*Habitable Worlds Observatory* [2]).

Mission yield estimates for HWO and LIFE typically follow one of two approaches: Either using partially theoretical, partially empirical occurrence rates to model exoplanet yields, or using the currently known exoplanet population for such predictions [3]. The latter may fall victim to observational biases in a stronger way, but presents the advantage of giving a concrete potential target list for the characterization phase, which may be important in building certain science cases for LIFE. This report builds on the latter approach.

In previous work, Menti et al. [4] with LIFEcat and Tuchow, Stark, and Mamajek [5] with HPIC (*Habitable Worlds Observatory Preliminary Input Catalog*) have compiled catalogs of stars based on physical criteria and mission-specific requirements. Specifically, HPIC filtered GAIA DR3 [6] and and the TESS Input Catalog [7] to stars with the distances less than 50 pc and TESS-Band and Gaia-Band magnitudes larger than 12, resulting in 12 944 remaining stars. On the other hand, LIFEcat used SIMBAD [8] as its main data source filtering in a more complex way based on temperature, age, flare activity and other criteria. Since the HWO and LIFE missions share many observation goals—and while LIFEcat is still in development—HPIC can be used to help with LIFE target selection. A current working version of a stellar catalog extends the HPIC catalog by adding stars in LIFEcat that are not contained in HPIC. This subset numbers 1627 stars and consists of M and K dwarfs cut off by the magnitude limits. The resulting catalog contains 14 571 stars, all of which have spectral classes F or later.

Crossmatching describes the process of identifying corresponding objects between two astronomical catalogs. This work focuses on finding the already detected planets around the HPICxLIFEcat stars, and provides a general Python package for matching a stellar catalog to a planetary catalog, outputting a combined catalog with combined columns. We take an approach that is agnostic to both the stellar catalog used, such that future iterations of a stellar target selection catalog can easily be processed, and to the planetary catalog used, such that different data sources may be swapped easily. The NASA Exoplanet Archive [9] and ExoMerCat [10, 11] are the catalogs this project was specifically developed for. In this project we differentiate between coordinate-based and identifier-based crossmatching and treat both.

1.2 Catalogs

The NASA Exoplanet Archive (NEA) is the most widely cited repository of confirmed exoplanets and their host-star properties. We make use of its `pscomppars` table. It aggregates and merges data from multiple published scientific sources to fill in missing gaps, prioritizing parameters based on measurement precision and recency, but also computes missing quantities. A breakdown of which columns have calculated values can be seen in fig-

ure 1. The methods by which the calculations are performed are documented in `https://exoplanetarchive.ipac.caltech.edu/docs/pscp_calc.html`, which this project makes use of at a later time. The archive also holds other tables with mission-specific exoplanet candidates, and records their promotion to confirmed planets, namely the TESS Objects of Interest Catalog [7] and the EPIC/K2 Objects of Interest Catalog [12], which are used indirectly through Exo-MerCat.

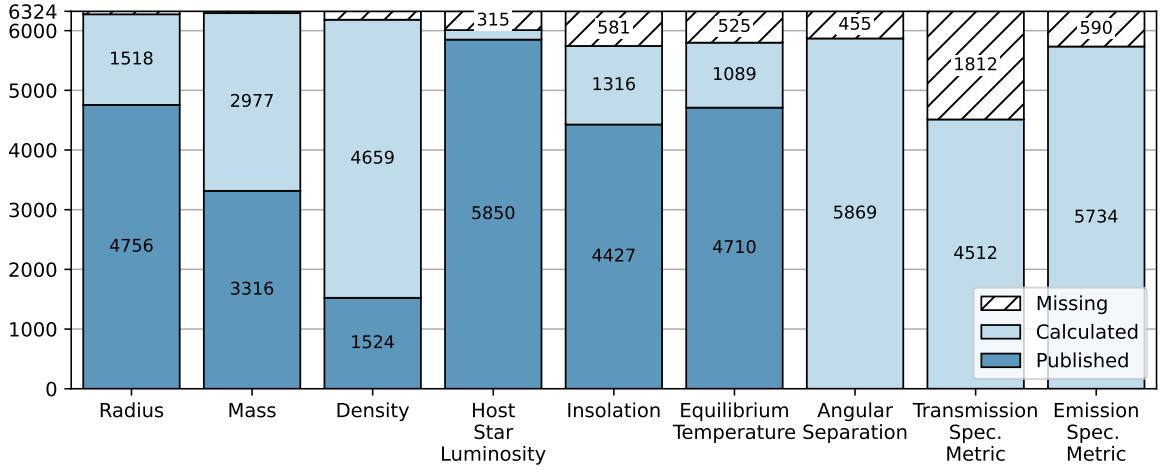


Figure 1: Provenance of `pscomppars` values (out of 6324 confirmed planets in total) for the nine columns in which NEA reports "Calculated Value" entries. All other columns are devoid of computations. Each bar splits the catalog into entries with a directly published measurement, entries calculated by the archive, and entries for which the parameter is missing entirely [9].

Exo-MerCat (*Exoplanet Merged Catalog*) is a Python tool designed to generate a catalog of exoplanets, where data is merged from the most prominent exoplanetary catalogs (NASA Exoplanet Archive [9], Exoplanets Encyclopaedia [13], Open Exoplanet Catalogue [14], the TESS Objects of Interest Catalog [7] and the EPIC/K2 Objects of Interest Catalog [12]). Exo-MerCat removes computed quantities from catalogs and chooses the most precise quantities from the catalog list. It creates a large table with homogenized column names and units. Exo-MerCat also saves the catalogs in which it finds the planets, since a given planet may appear in multiple catalogs. A breakdown of the sources in the form of a UpSet-plot can be seen in figure 2.

One important caveat that applies to both the NEA table and Exo-MerCat is that parameters for a given planet may not be self-consistent, as such planet-specific computations should be treated with caution [9, 10, 11]. We will explicitly calculate some missing parameters. Thus all results involving some calculation, which we always clearly mark, should be treated with caution.

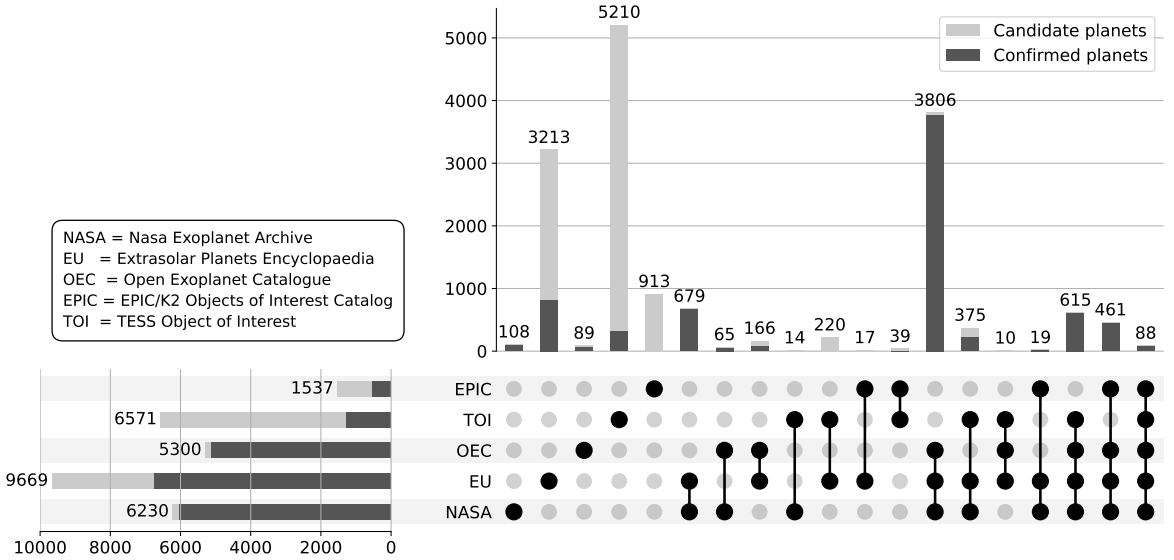


Figure 2: UpSet plot showing the overlap and distribution of exoplanets across the source catalogs of Exo-MerCat. The horizontal stacked bar chart on the left indicates the total count of planets in each catalog, while the vertical stacked bar chart at the top shows the size of the intersections of catalogs. Intersections with less than ten elements are omitted for brevity. A black dot on the line of a certain catalog means all planets in the vertical segment representing an intersection appear in that catalog. The bars are color-coded by planet status to differentiate between confirmed planets and candidate planets.

1.3 Conventions

A candidate exoplanet is a suspected planetary signal that requires further validation to rule out astrophysical false positives. Conversely, a confirmed exoplanet is a candidate whose planetary nature has been independently detected using complementary observational techniques, establishing a high degree of physical certainty. Some catalogs such as NASA Exoplanet Archive and Open Exoplanet Encyclopedia carry only confirmed planets[1], while others such TESS object of Interest Catalog and EPIC/K2 Candidate Planet Catalog carry both types. Exo-MerCat has its own internal rules on how to assign a planet status and this work, we accept their treatment of the matter and treat both kind of planets, but always distinguish between them.

Of particular interest are potentially habitable planets which are commonly defined as the planets able to host liquid water. We will be working with the conventions of the LIFE collaboration, defining habitable planets by being temperate in the sense that the insolation flux lies in the range of $0.35\,S_\oplus$ to $1.7\,S_\oplus$ [15] and that they may be rocky, which we define by the planetary radius being in the range of $0.5\,R_\oplus$ to $1.5\,R_\oplus$ [16] .

We distinguish two categories under which a planet can be classified as habitable. The first one is "Confirmed", when its published values lie within our definition. "Uncertain" means that the planet might be rocky and temperate in the sense that it is either a candidate planet

[1]The entries shown as candidate planets are ones that Exo-MerCat has flagged as "Controversial" since another catalog does not see them as confirmed

meeting the requirements or a planet whose measurement error in the relevant quantities makes the allowed parameter range overlap with our definition of habitable.

In section 3, we will repeatedly put planetary host stars into one of our categories, based on their spectral type:

- Very-low luminosity stars: M3V to M9V
- Low-luminosity stars: K6V to M2V
- Sun-like stars: F0V to K5V
- Other: all other main-sequence stars and non-main-sequence stars

While "other" in absolute terms encompasses the majority of known stars, in our samples the first three categories will dominate.

2 Methods

2.1 Python Package

The crossmatching and enrichment pipeline described in this work is implemented as an open-source Python package, available at `https://github.com/JoshuaDreier/ExoplanetCrossmatcher`. It makes extensive use of the package `astropy` [17, 18, 19], using `astropy.table.Table` as the primary data structure throughout, so that all intermediate and final outputs are directly compatible with the broader astronomical Python ecosystem. Astropy's connection to `pandas` [20] and `numpy` [21] was leveraged at multiple points in the code. Remote catalog access is handled via the Table Access Protocol (TAP) through `pyvo` (package associated with Astropy), with queries issued to the NASA Exoplanet Archive and SIMBAD endpoints. The package was developed and tested under Python 3.14. The package was written in a catalog-agnostic way, such that the planetary catalog as well as the stellar catalog are parameters of the code and can easily be changed. The same is true for the service supplying alternate identifiers. The NASA Exoplanet Archive and Exo-MerCat catalog were implemented and specifically developed for this project. As a source for alternate identifiers SIMBAD and Exo-MerCat were added. For instructions and details on the crossmatching and enrichment code, refer to the code documentation contained in the repository.

Using `pytest` [22], an end-to-end test suite uses the written code to match a set of known exoplanet host stars against all supported catalog and identifier-supplier combinations, asserting both that all expected planets are recovered and that no false positives are introduced. The identifier normalization rules described in section 2.2 were each developed by inspecting coordinate-matched planets that were not recovered by identifier matching, and were only retained after confirming they introduced no spurious matches in the test set. Furthermore, functional tests and unit tests were written alongside the crossmatching and enrichment code, double-checking most non-trivial functions for possible bugs, ensuring that human-made coding errors are mitigated before 'live-runs' yielding the actual data for this report.

2.2 Identifier-based crossmatching

Matching stellar catalogs by name is the most precise crossmatching strategy available: when the same physical object appears in both the input survey and the planet catalog under a common identifier, no position measurement is required and no false positives (FP) from chance alignments can arise (except for catalog-side wrong assignments of exoplanets to stars). The pipeline therefore prioritizes identifier-based matches over coordinate-based ones.

A key challenge in name-based catalog matching is that astronomical objects carry many designations: TIC numbers, HD/HIP/Gaia identifiers, common names, etc. none of which are guaranteed to appear in every catalog. To overcome this, the crossmatching queries the SIMBAD astronomical database [8] for every star in the input catalog, retrieving the full set of known identifiers for each object. For our input catalog, this produces a table of roughly 300 000 (input-star, alias) pairs. Exo-MerCat already provides an "alternate-identifiers" column, in which case the identifiers are taken from there and the SIMBAD query is omitted (Note that Exo-MerCat uses SIMBAD internally).

SIMBAD Identifier strings are not always directly compatible with how the stars are named in planetary catalogs. Therefore a few additional alternative identifiers are computed to increase

the number of correct matches. Here follows a list of the additional alternate identifiers added for each identifier of a certain type. This was done by manually looking of the sample of coordinate-matched but not ID-matched planets and inspecting their alternate identifiers. This way most naming inconsistencies in the catalog dataset became apparent and were adjusted for:

1. White-Space is normalized for all identifiers, some services store double spaces between catalog names and numbers (e.g. SIMBAD's `Ross  128`)
2. Object-type prefixes (`*` or `**`) are stripped (e.g. `* alf Cen` → `alf Cen`),
3. The `NAME` prefix is removed (e.g. `NAME Proxima Cen` → `Proxima Cen`)
4. Possessive suffixes are dropped (e.g. `Barnard's` → `Barnard`).

If the IDs are taken from Exo-MerCat other normalization rules are applied

1. White-space normalization
2. capitalizing the `Gaia` prefix to `GAIA`
3. TIC identifiers (`TIC-`N) to the space-separated form (`TIC` N)

All of these extra rules were verified to not produce extra false positives by comparing the entirety of the matched planets with a rule added to the same sample matched without that rule and confirming that the differently matched planets are close-enough in coordinates or in some cases manually inspecting them using SIMBAD to see if they are the same object.

In section 3 (Results) it is made clear that this is the most important matching method by number of matched planets.

Identifier crossmatching can also be used to detect and delete duplicates appearing, for example after merging parts of LIFEcat with HPIC.

2.3 Coordinate-based matching

Coordinate-based crossmatching serves as a complementary fallback for planets whose host star name could not be matched by any known identifier. For each row in the planet catalog, the closest star in the input catalog is found by angular distance, and the pair is accepted if that distance falls within a certain search radius r.

This is deliberately asymmetric. The search runs over the planet catalog, not the input catalog. This naturally handles multi-planet systems, such that all planets sharing a host star are each matched independently to the same input star.

The search radius is not the same for all planets. For catalog rows where the proper motion (PM) μ and its upper uncertainty σ_μ^+ are known, the radius grows with the expected positional drift between the planet catalog's coordinate epoch and the input catalog's epoch:

$$r = (\mu + \sigma_\mu^+)\,|\Delta t_i| + r_0 \tag{1}$$

where $r_0 = 10''$ is the base tolerance and Δt_i is the epoch difference in years. Since proper motion is typically assumed to be approximately uniform rectilinear motion [23], this formula describes an approximate maximum angular distance a star can have traveled in the time

between the two epochs. For rows where the proper motion or epoch is unknown, a set radius of $r_{\text{default}} = 50''$ is used instead. The size of r_0, r_{default} are calibrated empirically in section 3.1.1. With the current implementation, PM needs to be present as a property of the host star on the planet catalog side. Thus, only crossmatching against the NASA Exoplanet Archive can make use of the proper-motion-sensitive coordinate crossmatching, as it carries the PM values for its host star, which all other catalog sources of Exo-MerCat and Exo-MerCat itself do not. In principle however, with significant code adjustments, the matching algorithm could be written to take into account the PM on the stellar catalog side instead. As shown in the results section, the difference in accuracy is not large for our sample, because of the sparse density of exoplanet catalog entries at the time of writing.

The adopted tolerances are large compared to the $1''$ used in Exo-MerCat v2.1.0[2][11]. This is deliberate, as the appropriate value should be determined by the positional uncertainties of both catalogs, the epoch difference between them, and the on-sky density of the searched catalog. Since NEA does not publish per-row positional uncertainties and its coordinates are heterogeneously sourced (cf. section 3.1.1), the coordinate uncertainty term is replaced by the fixed base tolerance $10''$, chosen generously enough to absorb the worst coordinate entries (see section 3.1.1).

Such generous radii are only admissible because the searched catalog is sparse. HPICxLIFEcat contains roughly 10^4 stars spread over the full sky, corresponding to a surface density of $\rho \approx 2 \times 10^{-8}\,\text{arcsec}^{-2}$. Mathematically we can model the probability of a chance alignment $\mathbb{P}_{\text{fp}}$ for a single planet, by the likelihood of finding one star inside the circular search are around one planet, therefore $\mathbb{P}_{\text{fp},r} = \pi r^2 \cdot \rho$.[3] for the base tolerance and $5 \cdot 10^{-5}$ for the $50''$ fallback.
For the NEA strategy with proper motion aware radii forming a root mean squared[4] of 21.5", we get a per row FP chance of $\mathbb{P}_{\text{fp},21.5''} = 2.9 \cdot 10^{-5}$. With 6324 planets, this yields an expected FP count of 0.18. Note that the $r_{\text{default}} = 50''$ is chosen such that the chance of an FP stays just below one if we applied it to the entire dataset[5], which is a slightly arbitrary but fixed choice.
With the rougher Exo-MerCat matching technique of using only the set $r_{\text{default}} = 50''$ on 16132 planets yields 2.5 expected FPs.
However do note that this rough calculation A) assumed a uniform density of the exoplanet and in doing so B) neglected multiple planet systems. A correction of A) would likely increase FP chances, while with a correction of B) would likely decrease them. We must treat these computations as very rough calculations outlining that the chance of matching FPs is relatively low and luckily in section 3.1.1 we find no FP matches with either strategy.

2.4 Post-matching data enrichment

After crossmatching, many planet catalog rows are missing stellar or orbital parameters needed for candidate filtering or other uses. An enrichment step fills these gaps by merging values from additional catalog sources in a fixed priority order. The sources used here are, from highest to lowest priority: the input HPIC catalog, the NASA Exoplanet Archive

[2] Exo-MerCat v1.0.0 used search radii of up to 36"
[3] Giving $\mathbb{P}_{\text{fp},10''} = 2 \cdot 10^{-6}$ and $\mathbb{P}_{\text{fp},50''} = 5 \cdot 10^{-5}$
[4] a simple calculation shows that this is the correct way to achieve the average chance for a FP match
[5] works out to 99.3%

composite parameters table, the Exoplanets Encyclopaedia, the K2 EPIC catalog, the TESS Objects of Interest catalog, and SIMBAD stellar parameters. The first source that provides a non-masked value for a given parameter is taken as the entry for that quantity.

After sourcing, for still missing parameters, calculations are executed to give broad estimates for a subset of the sourced quantities. Each output parameter is accompanied by a provenance string that records which source or formula supplied the value.

The list of parameters that are sourced from catalogs are: planetary radius r_p, planetary mass m, projected planetary mass ($m \sin i$) planetary insolation flux S, planetary equilibrium temperature $T_{\rm eq}$, orbital semi-major axis a, orbital period P, host star spectral type (MK system), host star radius R_*, host star mass M_*, host star effective temperature $T_{\rm eff}$, host star surface gravity $\log g_*$, host star metallicity Fe/H, host star luminosity L_*, host system distance d and the magnitude brightness of the host star system in the Visual and Kepler bands.

For parameters that remain missing after source merging, a cascade of physically motivated fallback derivations is applied. Treating the variables as uncorrelated, gaussian error propagation is applied where possible to retain a sense of accuracy after the quantities have been computed. The derivations execute in the order listed below, each running only when its inputs are available and the target quantity is still missing.

- Host star effective temperature: Inferred from spectral type, interpolating between anchors from Pecaut and Mamajek [24] and Martins, Schaerer, and Hillier [25], or via Stefan Boltzmann relation $T_{\rm eff}^4 \propto \dfrac{L_*}{R_*}$
- Host star Luminosity: $L_\star \propto R_\star^2 T_{\rm eff}^4$ (Stefan-Boltzmann)
- Host star radius: The following options are tried, stopping at the first success, (methods decreasing in precision):
  - $R_*^2 \propto M_*/g_*$
  - $R_*^2 \propto \dfrac{L_*}{T_{\rm eff}^4}$ (Stefan Boltzmann)
  - Mann et al. [26] empirical relation dependent on the absolute K-band magnitude M_K, which is calculated from the K-band magnitude and distance (only applicable for $T_{\rm eff} < 4200\,\mathrm{K}$ and $M_K \in [4.0, 10.5]$). Then R_* follows from a second degree polynomial in M_K with a prefactor dependent on Fe/H.
  - Torres, Andersen, and Giménez [27] empirical relation applicable ($T_{\rm eff}$ from 3900 K to 8500 K), giving $\log R_*$ as a third degree polynomial in $\log T_{\rm eff}, \log g$, Fe/H.
  - Mann et al. [26] empirical relation for late K and M dwarfs ($T_{\rm eff} < 4000\,\mathrm{K}$), giving R_* as a third degree polynomial in $T_{\rm eff}$
  - Boyajian et al. [28, 29, 30] zero-age main-sequence relation between R_* and $T_{\rm eff}$, expressed as a piece-wise power law clamped at $0.05\,\mathrm{R}_\odot$
- Host star mass: $M_* \propto R_*^2 g_*$
- Orbital semi-major axis: $a^3 \propto M_* P^2$ (Kepler's third law)

- Planet Equilibrium Temperature: $T_{\mathrm{eq}}^4 \propto S$, (taking earth's eq. temperature with a Bond albedo of $A_B = 0.3$ as a reference.)
- Planet insolation: $S \propto L/a^2$ or via reversing equilibrium temperature formula above

Further calculation details can be found in the code documentation. A breakdown of the number of times each computation was employed for the dataset we are working with can be found in figure 10 in the appendix section A. The enrichment leads to $\sim 50\%$ more more planet entries in Exo-MerCat having insolation values that we can work with to characterize the population in section 3.

For radial-velocity planets with no directly measured radius, radius bounds are estimated from $m \sin i$ using the Chen & Kipping 2017 probabilistic mass-radius relation. The lower bound is $R_{\mathrm{CK}}(m \sin i)$ and the upper bound is $R_{\mathrm{CK}}(m \sin i / \sin i_{\mathrm{min}})$ with a default $\sin i_{\mathrm{min}} = 0.3$, based on averaged posterior-distribution estimates from Ho and Turner [32], giving a broad interval that keeps such planets visible in candidate-filtering workflows without asserting a definite radius. Similarly planet with known masses but unknown radius are given radius estimates with the same relation. This is the case for around 10% of all Exo-MerCat planets.

These enriched parameters are then used to classify planets as potentially rocky (r_p from $0.5\,\mathrm{R}_\oplus$ to $1.5\,\mathrm{R}_\oplus$) and potentially temperate (S from $0.35\,\mathrm{S}_\oplus$ to $1.7\,\mathrm{S}_\oplus$), described further in Section 3, in accordance with the definition given in section 1.3.

2.5 Exo-MerCat

At the time of writing, the Exo-MerCat [10, 11] TAP service is not actively updated. To guarantee up-to-date results, a local installation was used. The results in this report are based on Exo-MerCat v2.1.0 run on July 13, 2026, using the source catalogs most recent versions at that time.

3 Results

3.1 Matching

3.1.1 Coordinate Matching

The NASA exoplanet archive (NEA) is by far the most cited exoplanet catalog and thus deserves special attention among the input sources. NEA's coordinates are assembled from many different sources and the coordinate epoch can be read or estimated from the source they list for the entry. As explained in figure 3, using the proper-motion-aware equation (1) on a set of ID-matched stars, we achieve a matching accuracy of 100% (of the 792 ID-matched stars in the sample). In fact $r_0 = 10''$ is chosen as the lowest number for which this holds true. When using the simpler method of a fixed matching radius, then $r_{\text{min}} = 10''$ matches 81.4% of the sample and $r_{\text{fallback}} = 50''$ matches 99.4%. So, we would expect the same efficiency for 50" in the rows that the proper motion aware formula cannot be applied to, while keeping the FP chance low. By comparing for each planet which stars were matched, once by ID and once by coordinates, we note that none of the matching methods above introduced any FPs[6].

Exo-MerCat does not carry columns for proper motion, thus only a default search radius r_{default} can be applied. Comparing figure 3's median $4.1''$ crossmatching distance with figure 4's 3.0 mas, the average crossmatching distance between HPICxLIFEcat and Exo-MerCat is generally around three orders of magnitude smaller than the median crossmatching distance between HPICxLIFEcat and NEA. Due to this difference arising from the different coordinate sourcing between the two catalogs, we cannot naively transfer the search radii for the NEA method to EMC. However, $r_{\text{default}} = 50$ arcsec in this case is the minimal threshold such that all of the 1285 ID-matched ex-

[6] assuming the ID-matches to be false-positive free

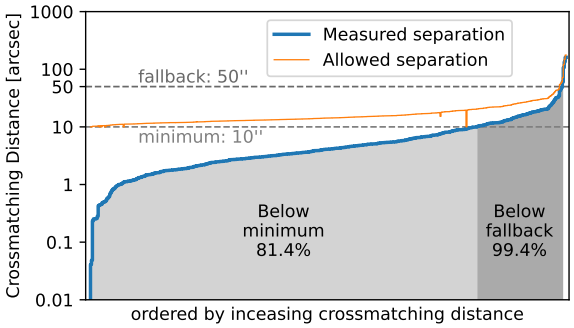


Figure 3: Calibration of coordinate crossmatching for NEA using SIMBAD IDs to crossmatch with HPICxLIFEcat. The separations between matched HPICxLIFEcat input stars and NEA catalog positions are plotted in increasing order and the algorithm's 10" minimum matching tolerance and 50" fallback threshold overlaid to illustrate how the observed IDs fall within the adopted angular acceptance. The remaining 0.6% above the fallback threshold are also coordinate matched, but only thanks to the proper-motion-sensitive search radius. The median crossmatching distance in this sample is $4.1''$.

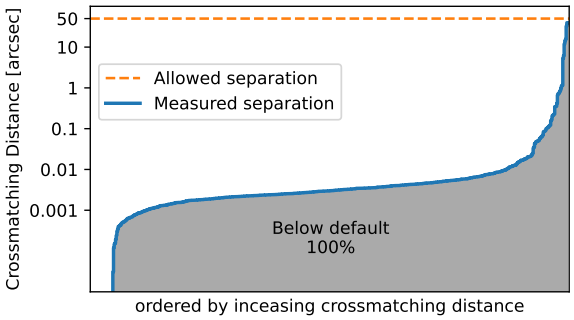


Figure 4: Calibration of coordinate crossmatching for EMC using ID crossmatches showing the measured angular separations for matched HPICxLIFEcat stars with the 50" matching tolerance annotated to demonstrate that all ID-based matches remain within the adopted acceptance radius. The median crossmatching distance in this sample is 3.0 mas.

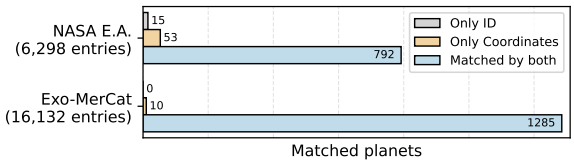


Figure 5: Yield of matched planets for the NASA Exoplanet Archive and Exo-MerCat catalogs, categorized by which methods the match was established.

oplanets in EMC can be matched via coordinates, with no FPs occuring.

3.1.2 Combined Matching

As shown in figure 5, matching HPICxLIFEcat with NEA produces a total of 860 matches, 92% of which were matched using both the coordinate and the ID method and 53 were only matched by coordinates, 15 solely by ID. Matching HPICxLIFEcat with Exo-MerCat, results in 1295 matches, with 99% of matches being made with both methods, only 10 coordinate-only matches and none being matched just by ID. Thus coordinate-only matching proves to be the edge case, as matching with NEA and Exo-MerCat primarily results in matches based on linked identifiers *and* coordinates. Exo-MerCat matches are almost a superset of the NEA matches; the only exceptions are high-mass directly imaged planets that were removed by Exo-MerCat's Brown Dwarf-removal step. However, note that despite Exo-MerCat being a superset of NEA, the absolute number of ID- and coordinate-only matches is lower for Exo-MerCat.

3.2 Population Analysis

Exo-MerCat assembles the largest published list of exoplanets, and is thus very useful in gaining a sense of the broader known planet population. In figure 6 we can see certain features of the diagram correspond to specific parts of the exoplanet population. Note that part of the data was taken from post-Exo-MerCat enrichment computations (a version with only published parameters can be found in figure 11 in Appendix section B): The "bulge" on the top right (high insolation, high radius) represents the hot jupiter type planets. The long thin strand pointing downwards on the right half of the plot are wide-separation planets observed via direct imaging.

We can compare the overall distribution with just the distribution of planets matched by HPICxLIFEcat, here done in heatmap form to better assess how many planets share similar parameters, presented in figure 7 (and a decomposition into the spectral categories of the same heatmap in appendix section C figure 12). Note some differences between the planetary populations populations:

- The feature of hot jupiters (upper right quadrant) is nearly absent in the HPICxLIFEcat planet population.
- The average radius and insolation values in the HPICxLIFEcat side are significantly lower than in the entire Exo-MerCat catalog ($4.4\,R_\oplus$ vs $7.5\,R_\oplus$ and $90\,S_\oplus$ vs $558\,S_\oplus$).

In contrast, the corresponding host star population mainly changes by an over-representation of late type stars[7] in the HPICxLIFEcat dataset.

Note further, in neither population are any planets excluded from being classified as temperate and rocky by having suitable insolation values and radii of less than $0.5R_\oplus$.

[7] here identified with "Very-low luminosity or Low-luminosity star"

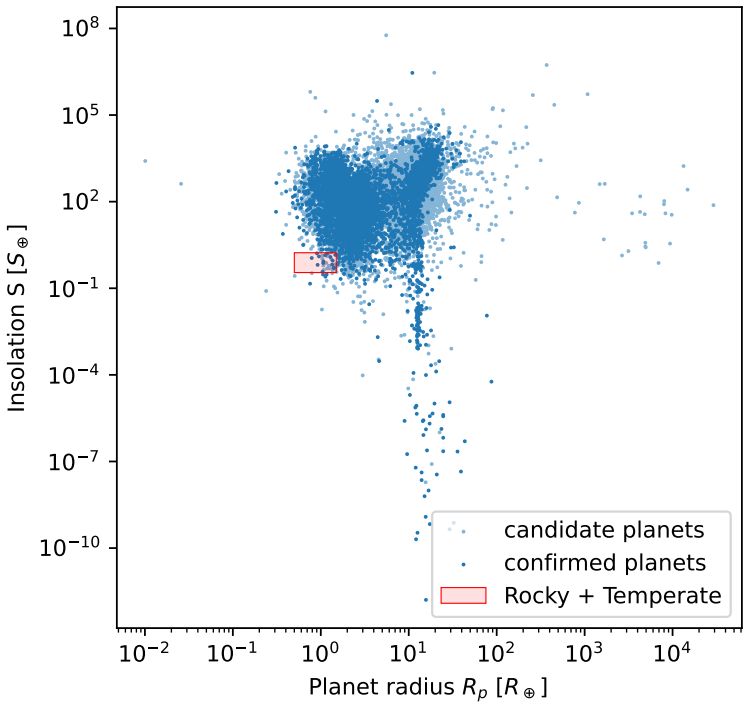


Figure 6: Distribution of the enriched Exo-MerCat exoplanet population in terms of planet-radius insolation (contains only published parameters). Confirmed planets are drawn in the darker shade and unconfirmed candidates in the lighter shade. The red rectangle marks the rocky-and-temperate selection region adopted in this work, spanning a radius of 0.5–1.5 $R_\oplus$ and an incident stellar flux of 0.35–1.7 $S_\oplus$.

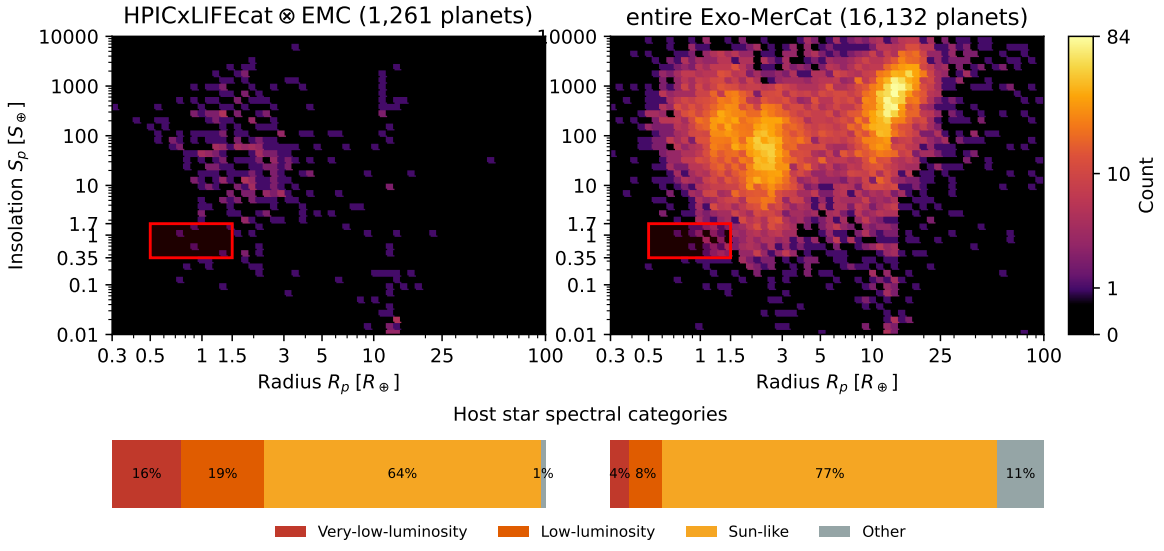


Figure 7: Two-dimensional number density heatmap of the known exoplanet population w.r.t planet-radius and insolation comparing the distributions in Exo-MerCat and Exo-MerCat matched to HPICxLIFEcat. The red rectangle marks the same rocky-and-temperate selection region as in figure 11, spanning 0.5–1.5 $R_\oplus$ and 0.35–1.7 $S_\oplus$. The distribution of the host star spectral categories as defined in section 1.3 is given in a stacked bar plot below the heatmaps. A breakdown of the distribution by host spectral category is given in figure 12 in Appendix section C.

Particularly interesting to the LIFE mission is the population of rocky habitable zone planets. Using the spectral categories and habitability conventions from section 1.3, we find the amount of habitable planets per spectral category for the general population and the crossmatched HPICxLIFEcat population in figure 8. The full list of the 20 (resp. 83) rocky, temperate planet candidates underlying that figure is given in section D: table 1 for the distance-limited HPIC×LIFEcat crossmatched sample and table 2 for the entire Exo-MerCat catalog.

Note that the share of habitable planets is significantly higher in the matched sample than in the general population: A total 20 habitable planets occupy 1.6% (7 confirmed, 0.5%) among 1216 in the crossmatched population; compared to 83 habitable planets (18 confirmed) among 16 132 planets in the general population giving 0.68% (0.1% confirmed).

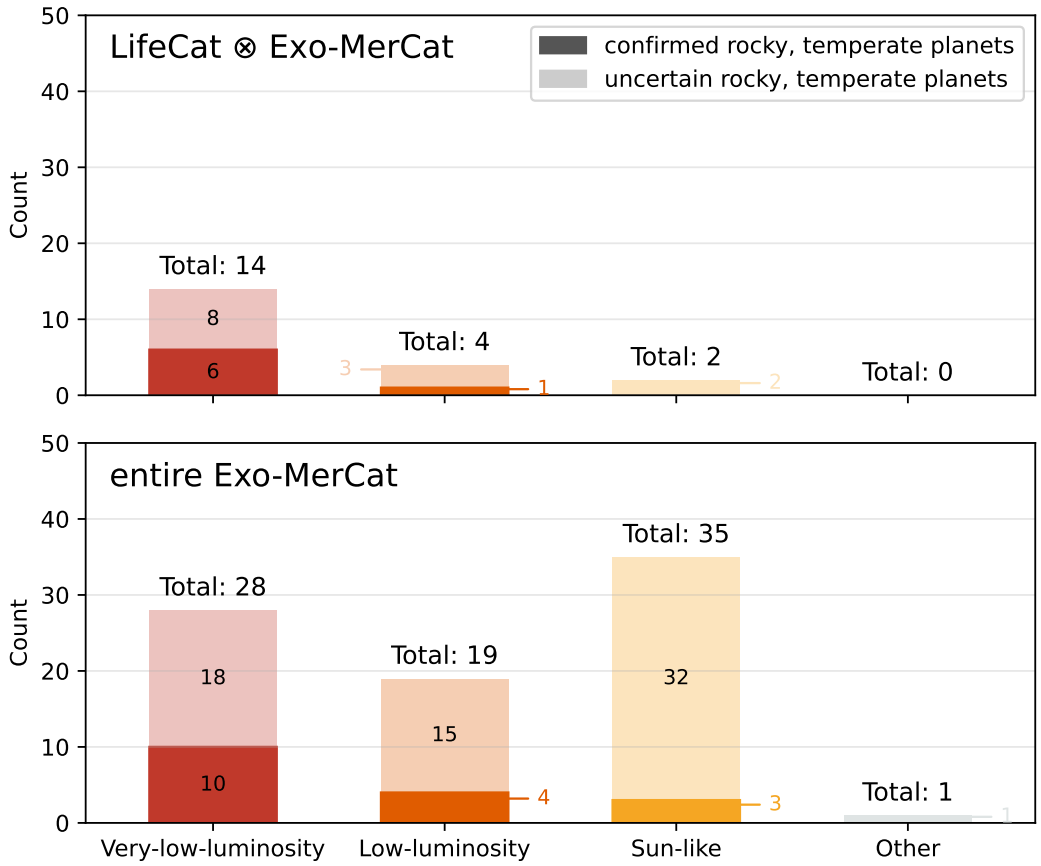


Figure 8: Yield of rocky, temperate planet candidates by host-star spectral category, using the same selection region as figure 11 and figure 7 (radius 0.5–1.5 $R_\oplus$, insolation 0.35–1.7 $S_\oplus$). A planet is counted as confirmed (dark shading) if its central-value radius and insolation both fall inside the box and it is a confirmed planet, and as uncertain if the box is met only within the uncertainty of the radius/insolation uncertainty interval, or if the central values fall inside the box but the planet is not confirmed. The top panel shows the distance-limited HPIC×LIFEcat sample crossmatched with Exo-MerCat (20 candidates in total), the bottom panel the entire Exo-MerCat catalog (83 candidates in total). The full list of candidates is given in Appendix section D.

4 Discussion

4.1 Crossmatching

Evidently from section 3.1.1, the crossmatching of HPICxLIFEcat with either NEA or EMC should be understood as successful and can be reasonably viewed as very close to perfectly reliable (i.e. bearing no false positives (FPs)) and complete (i.e. missing no planets)[8]. Furthermore, the crossmatching is easily reproducible and update-able on any changes to NEA, EMC or HPICxLIFEcat.

However, as mentioned in section 2.3, the coordinate-crossmatching radii of $r_0 = 10'', r_{\text{default}} = 50''$ is orders of magnitudes larger than other crossmatching programs. By the rough calculation in that section we can argue, the lower the search radii, the lower the expected chance of FPs coordinate matching, but the radii were chosen to be the minimal ones yielding completeness. Multiple ways of achieving smaller search radii while keeping completeness are imaginable: Using the similar coordinate epochs (lower Δt), which is of note in NEA as there the epochs of some entries lie in the 1950s. For EMC, a knowledge of the proper motion of the host stars would allow significantly smaller search radii, but that is a property that is currently not tracked.

Note also that the choice of $r_{\text{default}} = 50''$ currently is on poor footing regardless of its size, as our rough theoretical calculations do not prefer it over any value on the same order of magnitude. and the crossmatching methods still work with similar values (40", 60") without producing false positives. Currently the only advantage is that 50" coincidentally produces completeness and reliability for both NEA and EMC, but that is not to say there could be other, catalog dependent choices of r_{default}, that hold up better for higher on-sky densities. With, ID-matching dominating its precise value is not of much importance, however if having a coordinate-only catalog at hand or similar situation, this is a quantity that should be looked at in significantly more detail.

Further, for planetary catalogs, the technique could be improved (in the sense of yielding lower FP chances) on the side of this program by matching in a small area around an offset (by proper motion) position instead of the current method of tracing a circle with a radius determined by the proper motion as sketched in figure 9.

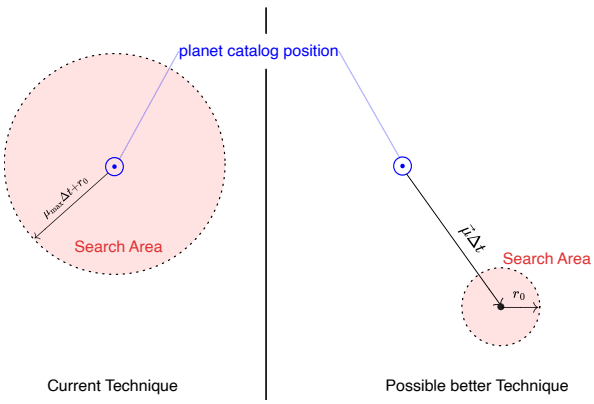


Figure 9: Sketch of a proposed improved technique of coordinate matching for planetary catalogs (NEA falls into this category, but not EMC)

Note the difference in average crossmatching distances in section 3.1.1 likely relates to the fact that Exo-MerCat takes its coordinates from SIMBAD, and so does LIFEcat, which informs part of the HPICxLIFEcat dataset. Further, HPIC's coordinates are sourced from TIC v8 [5] and Gaia DR3, the latter of which is also used to compute some of SIMBADs entries [8]. In opposition, NEA carries its own heterogeneously sourced coordinate entries, many of which do not even carry

[8] completeness was given in the ID-matched samples; we are extrapolating from that to the entire catalog.

the same epochs and sometimes reach back to the mid 20th century and are a priori unconnected to SIMBAD coordinates. For HPICxLIFEcat, this means that matching with EMC is preferred in view of long term avoidance of false positives. In the future, if a new catalog is assembled in a similar manner, ensuring the coordinate source (in the case of SIMBAD, also epochs) are chosen to match with the target planetary catalog (like EMC) will lower the needed search radius and thus keep the false positive occurrence rates low.

4.2 Parameter Enrichment

As mentioned in section 2.4, the computed parameters should generally be dealt with a great deal of caution. There are multiple reasons: firstly, the combination of differently sourced data might produce unphysical results, secondly, some of the computations involve empirical-statistical relations which can admit strong outliers. Still, for a broad overview of the population as aimed for in this report, they help to show general trends.

Comparing figure 11 (only published parameters) with figure 6 (includes computed parameters), we notice that the most extreme points come from candidate planets which typically have less trustable published parameters than confirmed planets. Furthermore, bunching appears around $10^1\,\mathrm{R}_\oplus$, which is a consequence of the Chen and Kipping [31] power law's transition around that value used to estimate radii when missing [9]. Since a large region of (minimum) mass space maps to the region close to $10^1\,\mathrm{R}_\oplus$ in radius space and many planets are known in that mass regime, they bunch around that value. By comparison of the two figures, we conclude the insolation flux values in that vertical line are mostly computed. They come only from directly imaged planets, which tend to have large separations (very low computed insolation) and are rarely published with insolation estimation. The exact location of these data points is thus to be treated with a high amount of skepticism, but their locations are again indicative of a general trend.

4.3 Population and its implication for LIFE

The change in host star population in section 3.2 between the Exo-MerCat and HPICxLIFEcat stars is qualitatively in agreement with the Malmquist bias [33], which is expected, as one of the main inclusion criteria for HPICxLIFEcat is a distance of less than 50 pc. More complicated selection criteria determine the HPIC and LIFEcat stars, but the distance cut is the dominant effect on the stellar host population in the broader sense. This macroscopic change in the host star population qualitatively explains the differences between the HPICxLIFEcat and Exo-MerCat planet populations seen in figure 7:

- The feature of hot Jupiters (upper right quadrant) is absent in the HPICxLIFEcat population; This is consistent with low-mass stars, which occupy a larger share of the host star population in HPICxLIFEcat, being known to only rarely host hot Jupiters [34].
- The shift to on average lower insolations and radii in the HPICxLIFEcat sample is consistent with observation bias against high periods and against low R_p/R_* ratios [35]. Therefore, lower mass (and thus radius) stars—which comprise a larger share of that sample compared to EMC—are more likely to have exoplanets with lower radii detected around them than their more massive counterparts.

- The share of habitable planets in the HPICxLIFEcat sample can be explained in the same manner

As this only touches on the most obvious features unrigorously, a deeper and more quantitative analysis of the differences of population could be covered in future work.

Note also that in either population, no planets are excluded from being classified as temperate and rocky by having suitable insolation values and less than $0.5R_{\oplus}$. This is likely due to observational biases against low-mass/radius and comparatively long-period planets [35], hinting that with better telescopes, those parts of phase space may feature more planets.

The population differences can be viewed with cautious optimism, as this means that the chosen set of stars is more likely than average to host known habitable exoplanets. This forms a strong argument for the chosen mission specifications, specifically wavelength ranges, for LIFE.

However, it would be somewhat premature to conclude that this means low-luminosity and very-low-luminosity stars host more habitable planets than their higher mass counterparts because, in general, it is not known which features of the known exoplanet population are attributable to observational bias and which are "fundamental". This disentanglement is the focus of statistical surveys that compare the observed population with the currently accepted theory [36]. Theoretical predictions of LIFE yield goes beyond the scope of this project, thus we only proclaim a better-than average habitability chance with respect to the *known* planet population.

4.4 Outlook

This project stands as a tool to quickly and reliably match known exoplanets to any stellar catalog and determine its habitable subpopulation, which can be used in the future for yield estimates similar to how a simulated exoplanetary catalog was employed by Kammerer, Quanz, and Dannert [16]. Furthermore, this work may be used for studies similar to "Large Interferometer For Exoplanets (LIFE): X. Detectability of currently known exoplanets and synergies with future IR/O/UV reflected-starlight imaging missions" by Carrión-González et al. [3] by providing a quick and repeatable way of selecting potentially habitable planets to then study on a case by case basis, which may inform the characterization phase target list.

Crossmatching performance, especially the certainty that we do not have false positive matches, could be improved by augmenting Exo-MerCat with a proper motion data sourcing or by implementing an algorithm taking into account the proper motion in the stellar catalog and by implementing more robust matching techniques as sketched in section 4.1 .

On the enrichment side, searching for more published empirical and physical relations of stars and planets could lead to more sophisticated enrichment pathways, which further increases the number of inferred parameters. Simultaneously, adding more data sources (for example, Gaia DR3 as done in [37]) could lead to an even better understanding of the broader known exoplanet population.

By adopting some of "Probing the limits of habitability: a catalogue of rocky exoplanets in the habitable zone" by Bohl et al. [37] methods—specifically the more intricate calculation of the habitable zones using polynomials with stellar temperature, including eccentricity in the

planet insolation considerations and their estimates of stellar age and its implication for the evolution of life—while using the larger dataset of Exo-MerCat could lead to a potentially different and more detailed understanding of the HPICxLIFEcat planet population or any catalogs habitable planet population for LIFE target selection.

5 Summary

In this work we presented an open-source, catalog-agnostic Python package that crossmatches a stellar catalog against a planetary catalog and enriches the matched planets with stellar and planetary parameters. Applied to HPICxLIFEcat, the pipeline was run against both the NASA Exoplanet Archive and Exo-MerCat, prioritizing identifier-based matching, using alternate identifiers from SIMBAD and Exo-MerCat, over a proper-motion-aware coordinate-based fallback.

Identifier-based matching proved to be the dominant matching method by number of matched planets, with coordinate-only matches remaining the edge case. Calibrating the coordinate thresholds on ID-matched stars showed that the proper-motion-aware search radius with a base tolerance of $10''$ recovers all ID-matches against NEA, and that the flat $50''$ fallback radius is sufficient when proper motion is unavailable; for Exo-MerCat $50''$ is the minimal search radius to achieve complete matching in the ID-matched sample. In both cases the sparse on-sky density of exoplanet hosts leads to a low number of predicted false positives and none found when coordinate matching the ID-matched sample. The Exo-MerCat matches form a near superset of the NEA matches.

After enrichment from a priority-ordered set of catalog sources and physically motivated fallback derivations, the combined catalog yields 20 rocky, temperate planet candidates around HPICxLIFEcat stars (7 confirmed and 13 uncertain), out of 83 across the entire Exo-MerCat catalog. This is a larger share than in the general exoplanet population, which can qualitatively be explained by observational biases related with the change in host star distribution. This list provides a set of already-known targets relevant to LIFE's search phase, and, owing to the catalog-agnostic design of the package, can be regenerated quickly as the stellar target-selection catalog and the planetary catalogs evolve.

6 Acknowledgments

The author thanks his advisor Dr. Lia Sartori for their guidance, continuous support and prompt responses, as well as his friends, Pascal D'Andrea and Johanna Kroker for proofreading and helpful discussions.

The author acknowledges using generative AI (Claude Code, Google Gemini) for the purpose of aiding in formatting the LaTeX-document, `matplotlib.pyplot` plots and as a search tool for finding relevant literature. The python package was extensively tested with AI-written testcases, the relevance and truthfulness of which was always verified by the author. The code-documentation follows the same principle. The author notes that no AI tools were used in the textual writing of the document (with the exception of checking for typos, and converting the summary into an abstract) and in the core logic of the code, excluding the aforementioned test cases.

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A Enrichment Pipeline

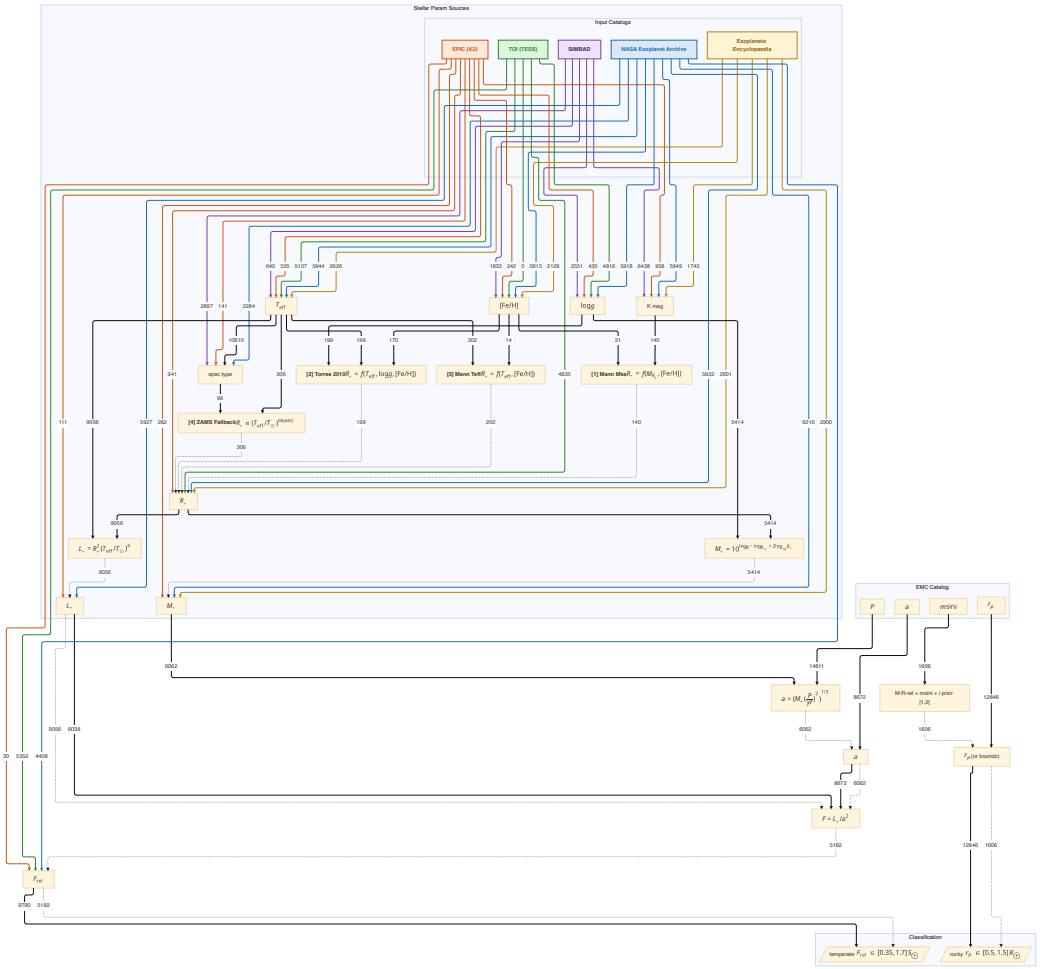


Figure 10: The flowchart illustrates the priority-ordered data retrieval hierarchy from input sources and the subsequent empirical or physical derivation pathways used to determine stellar parameters and orbital/planetary quantities. Solid arrows represent directly retrieved catalog values, while dashed arrows denote computed properties, with labels indicating the exact number of planet-rows processed through each transition.

B Planet distribution with only published parameters

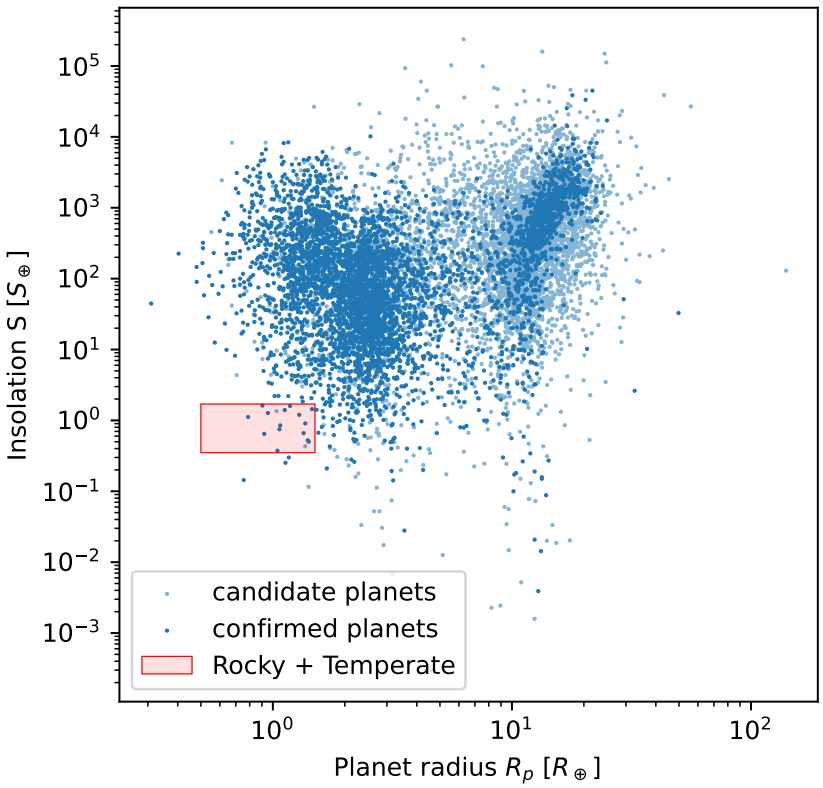


Figure 11: Distribution of the enriched Exo-MerCat exoplanet population in terms of planet-radius insolation (contains parameter computations). Confirmed planets are drawn in the darker shade and unconfirmed candidates in the lighter shade. The red rectangle marks the rocky-and-temperate selection region adopted in this work, spanning a radius of 0.5–1.5 $R_\oplus$ and an incident stellar flux of 0.35–1.7 $S_\oplus$.

.

C Population by Host Spectral Category

Figure 12 splits the planet-radius–insolation density of figure 7 by the spectral category assigned to the host star by the enrichment pipeline. As expected from the broader field, Sun-like hosts dominate the overall Exo-MerCat population. In the distance-limited HPICxLIFEcat crossmatch, however, the low- and very-low-luminosity categories carry a markedly larger share of the sample (∼35% of the crossmatched planets versus ∼12% of the full catalog), reflecting the abundance of cool, low-mass stars in the immediate solar neighborhood that dominates the LIFE target list.

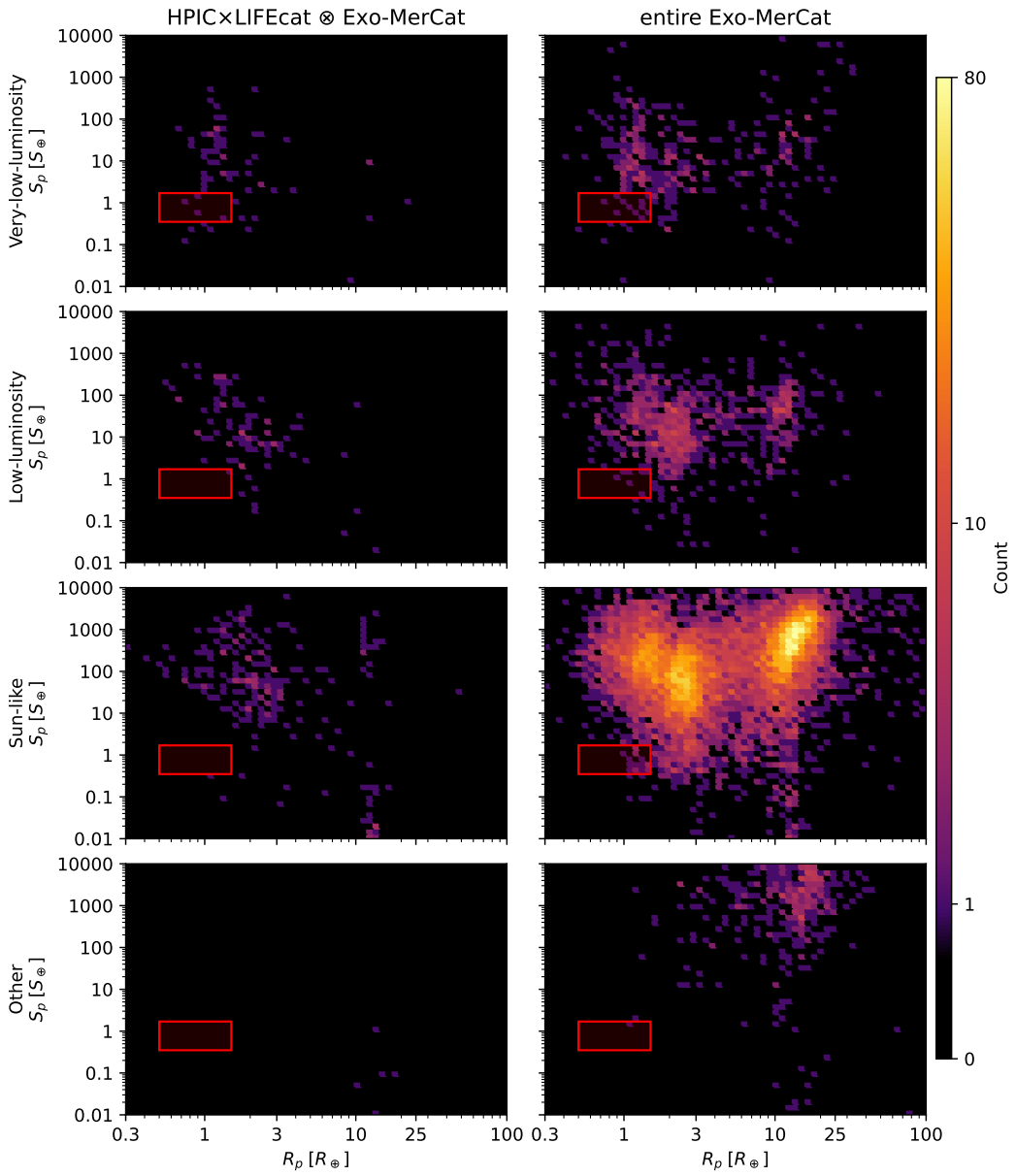


Figure 12: Two-dimensional number density of the exoplanet population in the planet-radius–insolation plane (as in figure 7), split by the spectral category of the host star. The four rows correspond to the host categories assigned by the enrichment pipeline (Very-low-luminosity, Low-luminosity, Sun-like, and Other); the left column is the distance-limited HPIC×LIFEcat sample crossmatched with Exo-MerCat and the right column the entire Exo-MerCat catalog. Both axes are logarithmic and the color encodes the number of planets per bin on a logarithmic scale, shared across all panels. The red rectangle marks the rocky-and-temperate selection region (0.5–1.5 $R_\oplus$, 0.35–1.7 $S_\oplus$). Planet radii and insolations are those provided by the enrichment pipeline.

D Rocky, Temperate Planet Candidate Tables

Table 1 lists the rocky, temperate planet candidates in the distance-limited HPIC×LIFEcat sample crossmatched with Exo-MerCat (the same population as the top panel of figure 8); table 2 lists the same selection applied to the entire enriched Exo-MerCat catalog (the bottom panel of figure 8).

Table 1: Rocky, temperate planet candidates in HPICxLIFEcat sample crossmatched with Exo-MerCat (see figure 8)

Planet	**Insolation Flux [$S_\oplus$]**	**Radius [$R_\oplus$]**	**Mass [$M_\oplus$]**	**Distance [pc]**	**Semi-major Axis [au]**	**Spectral Type**
			Confirmed			
L 372-58 d	0.60 ± 0.10	1.149 ± 0.076[c]	$1.67^{+0.17}_{-0.21}$	3.67	0.0540 ± 0.0010	M5.5
L 372-58 c	1.40 ± 0.20	1.194 ± 0.047[c]	$1.81^{+0.13}_{-0.11}$	3.67	0.0350 ± 0.0010	M5.5
G 32-5 b	$1.62^{+0.13}_{-0.11}$	$0.904^{+0.037}_{-0.034}$	0.71 ± 0.12	12.2	$0.067\,00^{+0.000\,50}_{-0.000\,60}$	M4
TRAPPIST-1 d	1.115 ± 0.043	$0.7880^{+0.0110}_{-0.0100}$	0.388 ± 0.012	12.5	0.0223	M7.5
TRAPPIST-1 e	0.646 ± 0.025	$0.920^{+0.013}_{-0.012}$	0.692 ± 0.022	12.5	0.0293	M7.5
TRAPPIST-1 f	$0.373^{+0.015}_{-0.014}$	$1.045^{+0.013}_{-0.012}$	1.039 ± 0.031	12.5	0.0385	M7.5
K2-3 d	1.44 ± 0.14	$1.458^{+0.056}_{-0.051}$	2.20	43.9	$0.2014^{+0.0034}_{-0.0033}$	M1
			Uncertain			
Proxima Centauri b	0.641 ± 0.065	1.36 ± 0.34[c]	1.055 ± 0.055[b]	1.30	$0.048\,48 \pm 0.000\,29$	M5.5
Ross 128 b	1.38	1.55 ± 0.45[c]	1.40 ± 0.21[b]	3.37	0.0496 ± 0.0017	M4
G 158-27 b	0.670 ± 0.039	1.37 ± 0.34[c]	1.08 ± 0.13[b]	4.85	0.0457 ± 0.0012	M5.5
BD-17 588 A d	1.10 ± 0.49[a]	2.21 ± 0.75[c]	2.721 ± 0.075[b]	6.86	0.090 ± 0.020	M3.0
HD 156384 C f[d]	$0.56^{+0.13}_{-0.11}$[a]	2.12 ± 0.72[c]	2.5 ± 1.3[b]	7.24	$0.156^{+0.014}_{-0.017}$	M1.5
HD 156384 C e[d]	0.302 ± 0.060[a]	2.20 ± 0.75[c]	$2.7^{+1.6}_{-1.4}$[b]	7.24	0.213 ± 0.020	M1.5
G 192-13 b	0.910 ± 0.090	2.00 ± 0.68[c]	2.30 ± 0.40[b]	7.73	$0.096\,73 \pm 0.000\,80$	M4.0
Wolf 1069 b	$0.652^{+0.029}_{-0.027}$	1.48 ± 0.40[c]	1.26 ± 0.21[b]	9.57	0.0672 ± 0.0014	M5.0
G 261-6 b	$1.80^{+0.12}_{-0.11}$	1.54 ± 0.44[c]	1.37 ± 0.23[b]	10.6	$0.029\,71^{+0.000\,87}_{-0.000\,93}$	M6
HD 6660 .01[d]	1.39	2.24 ± 0.92	—	20.5	1.87[a]	K4
CD-34 1169 .01[d]	0.431	1.37 ± 0.77	—	25.1	1.17[a]	M
MCC 189 .01[d]	0.406	1.74 ± 0.84	—	39.9	1.51[a]	M0
HD 137010 b	$0.294^{+0.067}_{-0.200}$[a]	1.060 ± 0.061	—	44.8	$0.88^{+0.30}_{-0.10}$	K3.5

[a] value derived by the enrichment pipeline rather than directly measured; [b] minimum mass ($M \sin i$) rather than true mass; [c] radius estimated from mass or $M \sin i$ (not directly measured); [d] Only a candidate planet.

Table 2: Rocky, temperate planet candidates in Exo-MerCat (see figure 8)

Planet	**Insolation Flux [$S_\oplus$]**	**Radius [$R_\oplus$]**	**Mass [$M_\oplus$]**	**Distance [pc]**	**Semi-major Axis [au]**	**Spectral Type**
			Confirmed			
L 372-58 c	1.40 ± 0.20	1.194 ± 0.047^{c}	$1.81^{+0.13}_{-0.11}$	$3.672\,78^{+0.000\,89}_{-0.000\,90}$	0.0350 ± 0.0010	M5.5
L 372-58 d	0.60 ± 0.10	1.149 ± 0.076^{c}	$1.67^{+0.17}_{-0.21}$	$3.672\,78^{+0.000\,89}_{-0.000\,90}$	0.0540 ± 0.0010	M5.5
G 32-5 b	$1.62^{+0.13}_{-0.11}$	$0.904^{+0.037}_{-0.034}$	0.71 ± 0.12	12.210 ± 0.013	$0.067\,00^{+0.000\,50}_{-0.000\,60}$	M3.0
TRAPPIST-1 d	1.115 ± 0.043	$0.7880^{+0.0110}_{-0.0100}$	0.388 ± 0.012	12.430 ± 0.019	0.0223	M8.0
TRAPPIST-1 e	0.646 ± 0.025	$0.920^{+0.013}_{-0.012}$	0.692 ± 0.022	12.430 ± 0.019	0.0293	M8.0
TRAPPIST-1 f	$0.373^{+0.015}_{-0.014}$	$1.045^{+0.013}_{-0.012}$	1.039 ± 0.031	12.430 ± 0.019	0.0385	M8.0
TOI-700 d	$0.850^{+0.090}_{-0.100}$	$1.073^{+0.059}_{-0.054}$	2.42	31.127 ± 0.021	0.1633 ± 0.0027	M2.5
TOI-700 e	$1.27^{+0.13}_{-0.15}$	$0.953^{+0.089}_{-0.075}$	—	31.127 ± 0.021	0.1340 ± 0.0022	M2.5
LP 890-9 c	0.906 ± 0.026	$1.367^{+0.055}_{-0.039}$	25 ± 25	32.430 ± 0.070	$0.039\,84 \pm 0.000\,22$	M6
K2-3 d	1.44 ± 0.14	$1.458^{+0.056}_{-0.051}$	2.20	44.07 ± 0.11	$0.2014^{+0.0034}_{-0.0033}$	M
LP 759-72 e	1.20 ± 0.20	1.29 ± 0.13	—	66.43 ± 0.20	$0.1060^{+0.0090}_{-0.0130}$	—
Kepler-1649 A c	$0.750^{+0.032}_{-0.302}$	$1.06^{+0.15}_{-0.10}$	—	$92.19^{+0.55}_{-0.54}$	0.0649	—
Kepler-1512 A b	$1.60^{+0.44}_{-0.51}$	$1.180^{+0.070}_{-0.110}$	—	$162.0^{+5.8}_{-15.9}$	$0.136\,93 \pm 0.000\,52$	—
Kepler-438 b	$1.40^{+0.67}_{-0.77}$	$1.12^{+0.16}_{-0.17}$	—	195.9 ± 4.7	$0.166^{+0.051}_{-0.042}$	—
Kepler-1229 A b	$0.524^{+0.054}_{-0.050}$	$1.40^{+0.11}_{-0.13}$	2.54	$265.5^{+3.4}_{-3.3}$	$0.3006^{+0.0069}_{-0.0091}$	M0
Kepler-62 f	0.500	1.412 ± 0.067	35.0	300.9 ± 1.2	0.7180 ± 0.0070	K2
Kepler-442 b	$0.66^{+0.23}_{-0.41}$	$1.35^{+0.18}_{-0.11}$	—	$366.0^{+2.8}_{-2.7}$	$0.409^{+0.209}_{-0.060}$	—
			Uncertain			
Proxima Centauri b	0.641 ± 0.065	1.36 ± 0.34^{c}	1.055 ± 0.055^{b}	$1.301\,19^{+0.000\,34}_{-0.000\,35}$	$0.048\,48 \pm 0.000\,29$	M5.5
Ross 128 b	1.38	1.55 ± 0.45^{c}	1.40 ± 0.21^{b}	$3.374\,54 \pm 0.000\,80$	0.0496 ± 0.0017	M4
Teegarden's Star b	1.080 ± 0.080	1.42 ± 0.37^{c}	$1.16^{+0.12b}_{-0.11}$	3.8308 ± 0.0040	$0.025\,90^{+0.000\,80}_{-0.000\,90}$	M7.0
Teegarden's Star c	0.350 ± 0.020	1.36 ± 0.33^{c}	1.05 ± 0.14^{b}	3.8308 ± 0.0040	$0.0443^{+0.0014}_{-0.0015}$	M7.0

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Table 2: Rocky, temperate planet candidates in Exo-MerCat (see figure 8) (Continued)

Planet	**Insolation Flux [$S_\oplus$]**	**Radius [$R_\oplus$]**	**Mass [$M_\oplus$]**	**Distance [pc]**	**Semi-major Axis [au]**	**Spectral Type**
G 158-27 b	0.670 ± 0.039	1.37 ± 0.34[c]	1.08 ± 0.13[b]	4.8487 ± 0.0030	0.0457 ± 0.0012	M5.5
BD-17 588 A d	1.00 ± 0.45[a]	2.21 ± 0.75[c]	2.721 ± 0.075[b]	6.90	0.090 ± 0.020	M3.5
HD 156384 C e[d]	0.302 ± 0.060[a]	2.20 ± 0.75[c]	$2.7^{+1.6}_{-1.4}$[b]	$7.2440^{+0.0049}_{-0.0048}$	0.213 ± 0.020	M1.5
HD 156384 C f[d]	$0.56^{+0.13}_{-0.11}$[a]	2.12 ± 0.72[c]	2.5 ± 1.3[b]	$7.2440^{+0.0049}_{-0.0048}$	$0.156^{+0.014}_{-0.017}$	M1.5
G 192-13 b	0.910 ± 0.090	2.00 ± 0.68[c]	2.30 ± 0.40[b]	7.7227 ± 0.0048	$0.096\,73 \pm 0.000\,80$	M4
Wolf 1069 b	$0.652^{+0.029}_{-0.027}$	1.48 ± 0.40[c]	1.26 ± 0.21[b]	9.5834 ± 0.0048	0.0672 ± 0.0014	M5.0
G 261-6 b	$1.80^{+0.12}_{-0.11}$	1.54 ± 0.44[c]	1.37 ± 0.23[b]	10.6267 ± 0.0063	$0.029\,71^{+0.000\,87}_{-0.000\,93}$	M5.5
L 291-115 b[d]	0.332 ± 0.023[a]	0.79 ± 0.17	—	14.6058 ± 0.0061	0.0699 ± 0.0014[a]	M5
HD 6660 .01[d]	1.39	2.24 ± 0.92	—	20.527 ± 0.029	1.87 ± 0.14[a]	K4
TOI-4336 .01	1.69	2.05 ± 0.76	—	22.546 ± 0.038	0.0850 ± 0.0017[a]	M3.5
CD-34 1169 .01[d]	0.431	1.37 ± 0.77	—	25.143 ± 0.030	1.176 ± 0.024[a]	M
LP 873-20 b[d]	1.54[a]	1.16 ± 0.25	—	26.075 ± 0.022	—	—
2MASS J06034353-8127303 .01[d]	1.35	1.04 ± 0.17	—	26.741 ± 0.063	0.0258 ± 0.0013[a]	—
G 125-49 b	$1.68^{+0.17}_{-0.37}$	$1.65^{+0.22}_{-0.20}$	—	32.027 ± 0.045	$0.057\,22 \pm 0.000\,44$	M3.5
LSPM J1957+5658 .01[d]	0.382	1.81 ± 0.66	—	32.609 ± 0.044	—	M4.0
PM J13119+6550 d	$1.30^{+0.30}_{-0.20}$	1.70 ± 0.58[c]	$3.7^{+1.0}_{-1.1}$[b]	36.012 ± 0.030	0.1513 ± 0.0024	M2
MCC 189 .01[d]	0.406	1.74 ± 0.84	—	39.889 ± 0.031	1.51 ± 0.16[a]	M0
LSPM J1902+7525 c	1.84 ± 0.17	1.334 ± 0.078	7.40	41.918 ± 0.050	0.1370 ± 0.0060	M2.5
TOI-715 b	$0.67^{+0.15}_{-0.20}$	1.550 ± 0.064	3.02	$42.405^{+0.086}_{-0.085}$	0.0830 ± 0.0027	M4
LP 425-186 .01[d]	35^{+32}_{-39}[a]	$1.26^{+0.39}_{-0.87}$	—	42.89 ± 0.18	$0.0117^{+0.0023}_{-0.0040}$[a]	—

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Table 2: Rocky, temperate planet candidates in Exo-MerCat (see figure 8) (Continued)

Planet	**Insolation Flux [$S_\oplus$]**	**Radius [$R_\oplus$]**	**Mass [$M_\oplus$]**	**Distance [pc]**	**Semi-major Axis [au]**	**Spectral Type**
HD 137010 b	$0.300^{+0.068a}_{-0.204}$	1.060 ± 0.061	—	44.860 ± 0.030	$0.88^{+0.30}_{-0.10}$	K3.5
2MASS J05021345+1442367 .01[d]	0.60 ± 0.22[a]	$1.233^{+0.116}_{-0.094}$	—	45.99 ± 0.72	$0.050\,61 \pm 0.000\,52$[a]	M4
2MASS J05021345+1442367 .02[d]	0.268 ± 0.098[a]	$0.505^{+0.088}_{-0.084}$	—	45.99 ± 0.72	$0.075\,52 \pm 0.000\,77$[a]	M4
2MASS J14441864+8426548 .01[d]	0.644	1.64 ± 0.90	—	61.761 ± 0.085	0.2979 ± 0.0062[a]	—
CD-64 238 .01[d]	0.836	2.09 ± 0.66	—	62.51 ± 0.14	1.52 ± 0.32[a]	—
Kepler-138 e	1.73 ± 0.12	0.82 ± 0.15[c]	$0.429^{+0.191}_{-0.095}$	66.86 ± 0.11	0.1803 ± 0.0014	M1
LP 613-39 b	$1.36^{+1.59}_{-0.81}$	$2.25^{+0.53}_{-0.96}$	—	82.96 ± 0.37	$0.091^{+0.013}_{-0.016}$	M2.5
HD 87794 .01[d]	0.467	2.3 ± 1.3	—	83.6	1.58	G3
BD+83 461 .01[d]	0.857	1.63 ± 0.67	—	99.90 ± 0.25	—	G0
BD+42 1620 b[d]	0.653	2.8 ± 1.5	—	110	1.62	G0
TYC 8897-1221-1 .01[d]	0.379	2.6 ± 1.1	—	159.05 ± 0.50	1.63 ± 0.10[a]	—
K2-296 b	$1.71^{+0.43a}_{-0.51}$	$1.87^{+0.20}_{-0.45}$	—	159.3 ± 1.4	$0.1346^{+0.0123a}_{-0.0053}$	—
Kepler-296 A e	$1.41^{+0.25}_{-0.21}$	$1.52^{+0.27}_{-0.25}$	—	167^{+21}_{-20}	$0.169^{+0.029}_{-0.028}$	M2
Kepler-296 A f	$0.62^{+0.11}_{-0.10}$	$1.80^{+0.31}_{-0.30}$	—	167^{+21}_{-20}	$0.255^{+0.043}_{-0.042}$	M2
Kepler-186 f	$0.30^{+0.10}_{-0.15}$	1.166 ± 0.078	—	177.59 ± 0.77	0.356 ± 0.048	M1
CPD-62 485 .03[d]	1.03	3.0 ± 1.9	—	193	1.59	—
CD-61 922 .01[d]	1.69	3.3 ± 2.2	—	232.3 ± 1.2	1.62 ± 0.11[a]	—

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Table 2: Rocky, temperate planet candidates in Exo-MerCat (see figure 8) (Continued)

Planet	**Insolation Flux [$S_\oplus$]**	**Radius [$R_\oplus$]**	**Mass [$M_\oplus$]**	**Distance [pc]**	**Semi-major Axis [au]**	**Spectral Type**
Kepler-1652 b	$0.81^{+0.16}_{-0.13}$	1.60 ± 0.18	—	252^{+32}_{-30}	$0.1654^{+0.0042}_{-0.0075}$	—
BD-06 3861 .01[d]	$209^{+981\text{a}}_{-329}$	$2.8^{+6.5}_{-1.5}$	—	$273.7^{+4.1}_{-4.0}$	$0.169^{+0.034\text{a}}_{-0.031}$	F2
KOI-5087 .01[d]	0.714[a]	$1.55^{+0.59}_{-0.35}$	—	307.13 ± 0.83	1.45	—
UCAC4 665-070369 b[d]	0.369 ± 0.070[a]	1.51 ± 0.13	—	330.9 ± 3.0	0.444 ± 0.040	—
KOI-4878 b[d]	1.05[a]	$1.04^{+0.38}_{-0.14}$	—	349.9 ± 1.1	1.14	G4
Kepler-1505 A .02[d]	0.483[a]	$1.18^{+0.44}_{-0.16}$	—	359	1.20	G0.5
KOI-7179 .01[d]	1.30[a]	$1.18^{+0.49}_{-0.22}$	—	397.0 ± 1.4	1.08	G2
KOI-6108 .01[d]	0.612[a]	$1.07^{+0.47}_{-0.16}$	—	408.7 ± 4.5	1.13	—
KOI-5806 .01[d]	1.29[a]	$1.26^{+0.49}_{-0.13}$	—	439.8 ± 1.7	0.898	—
KOI-5938 .01[d]	1.03[a]	$1.66^{+1.00}_{-0.22}$	—	538.9 ± 2.9	1.34	G0
KOI-5948 .01[d]	0.580[a]	$1.233^{+0.549}_{-0.080}$	—	539.1 ± 3.2	0.996	G5
Kepler-452 b	$1.10^{+0.29}_{-0.22}$	$1.63^{+0.23}_{-0.20}$	—	$551.7^{+5.2}_{-5.1}$	$1.046^{+0.019}_{-0.015}$	G2
Kepler-1126 c	$2.06^{+0.76}_{-0.59}$	1.45 ± 0.18	—	635.7 ± 6.8	0.619	—
KOI-5545 .01[d]	0.429[a]	$1.081^{+0.482}_{-0.069}$	—	678 ± 54	1.25	—
KOI-7235 .01[d]	0.752[a]	$1.155^{+0.381}_{-0.080}$	—	707 ± 293	0.825	—
Kepler-51 e	1.142 ± 0.040[a]	2.09 ± 0.74[c]	5.4 ± 1.5	784^{+14}_{-13}	0.795 ± 0.014[a]	—
KOI-5068 .01[d]	1.69[a]	$1.57^{+0.52}_{-0.24}$	—	806.2 ± 9.9	1.01	—
KOI-5819 .01[d]	0.483[a]	$1.244^{+0.415}_{-0.080}$	—	836 ± 13	0.955	—
KOI-5737 .01[d]	0.988[a]	$1.32^{+0.50}_{-0.13}$	—	875.0 ± 9.0	1.01	—
KOI-5499 .01[d]	1.43[a]	$1.19^{+0.15}_{-0.11}$	—	879 ± 17	0.419	—
KOI-7223 .01[d]	0.570[a]	$1.52^{+0.38}_{-0.12}$	—	986 ± 32	0.835	—
KOI-5506 .01[d]	0.659[a]	$1.61^{+0.80}_{-0.13}$	—	1006 ± 15	1.50	—

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Table 2: Rocky, temperate planet candidates in Exo-MerCat (see figure 8) (Continued)

Planet	**Insolation Flux [$S_\oplus$]**	**Radius [$R_\oplus$]**	**Mass [$M_\oplus$]**	**Distance [pc]**	**Semi-major Axis [au]**	**Spectral Type**
KOI-6425 .01[d]	0.677[a]	$1.46^{+0.57}_{-0.19}$	—	1120 ± 17	1.22	—
KOI-4986 .01[d]	0.468[a]	$1.480^{+0.471}_{-0.080}$	—	1323 ± 515	1.06	—
KOI-6151 .01[d]	0.466[a]	$1.356^{+0.471}_{-0.090}$	—	1547 ± 34	1.04	—